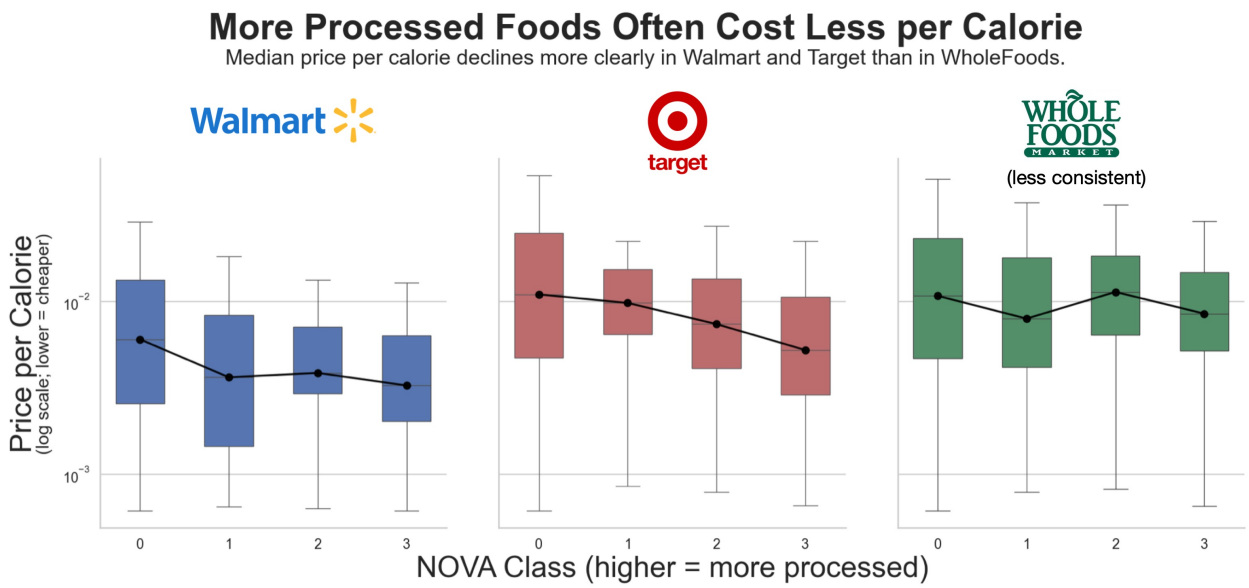

Notes: y-axis is log-scaled to compare prices spanning a wide range; lower values indicate cheaper calories.

Writeup

This visualization is meant to communicate one main takeaway: more processed foods often cost less per calorie, although that pattern is not equally strong across stores. I wanted the chart to show how price-per-calorie distributions change across NOVA processing classes while still making it easy to compare Walmart, Target, and WholeFoods. Instead of focusing on individual products, the figure highlights the broader relationship between processing level and calorie cost, since that is the main story I wanted readers to notice.

I used faceted boxplots because the question is really about comparing distributions across ordered categories. Boxplots work well here because they show median, spread, and overall variation in price per calorie within each NOVA class, which gives a fuller picture than plotting only averages. Splitting the stores into three separate panels reduces clutter and makes store-level comparisons easier than putting everything on one shared axis. I also added a black median trend line on top of each boxplot so the overall direction within each store is easier to see at a glance. This helps because the important comparison is not just between categories, but also between the shapes of the store-specific trends. I used one solid, brand-inspired color per store so that each panel keeps a clear store identity without adding unnecessary visual complexity.

Several transformations were important for making the comparison readable. I restricted the data to valid observations with non-missing values, positive price per calorie, and NOVA classes between 0 and 3. The dataset already provides FPro_class as a discrete predicted NOVA classification, so I used those classes directly rather than creating my own bins; converting the field to integer type simply helped preserve a clean category order in the plot. Because price per calorie varies across a wide range, I used a logarithmic y-axis so differences across classes could be compared without a few very large values dominating the figure. I also trimmed the plotted data to the 1st–99th percentile range for readability, while keeping the overall cleaned structure of the dataset intact. Finally, I fixed the panel order as Walmart, Target, and WholeFoods so the comparison is stable and easy to follow from left to right.

The design went through several iterations. Earlier versions either combined all stores in one plot or relied more heavily on noisier trend views, but those approaches made the main takeaway harder to read. Separating stores into facets and returning to boxplots produced a cleaner summary while still preserving distributional information. One limitation of this design is that it combines many different food categories, so category-specific patterns are hidden. In addition, WholeFoods does not follow the same steady decline as Walmart and Target, so the title intentionally uses “often” rather than suggesting a universal rule. Overall, the final design focuses on giving a clear and persuasive summary of the relationship between processing level and calorie cost while still showing where that relationship differs across stores.